# Customer Churn Analysis Dashboard

Data Science & Analytics internship – Task 2

Submitted BY :

Koya Devisri Chowdary

## Tools & Technologies Used

- Python
- Power BI
- Pandas
- Matplotlib
- Jupyter Notebook

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## PROJECT OVERVIEW

This project analyzes customer churn patterns in a telecom company using Python and Power BI. The main goal is to understand customer behavior, identify factors influencing churn, and create interactive visual dashboards for business decision-making.

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## PROJECT GOALS

- Study customer churn trends
- Identify major churn factors
- Visualize customer retention patterns
- Build an interactive Power BI dashboard
- Generate business recommendations

## DATASET INFORMATION

**Dataset Used:**

**Telco Customer Churn Dataset**

**Dataset Includes:**

- Customer demographic details
- Internet service information
- Contract types
- Payment methods
- Monthly and total charges
- Customer churn status

**Dataset Size:**

- 7043 Records
- 21 Attributes

## DATA CLEANING & PREPARATION

The dataset was preprocessed using Python:

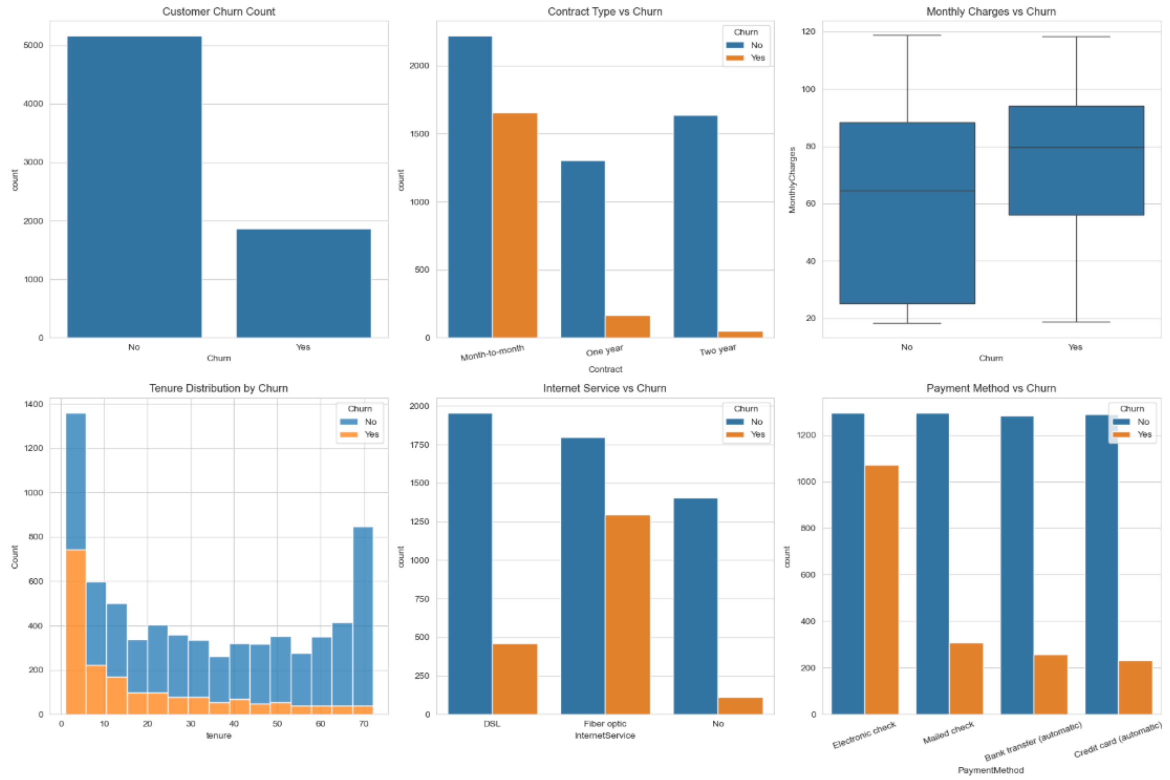
- Imported dataset using Pandas
- Converted data types for analysis
- Handled missing values
- Performed exploratory data analysis
- Prepared data for visualization

## PYTHON ANALYSIS PERFORMED

The following analyses were completed:

- Churn Distribution Analysis
- Contract Type Analysis
- Monthly Charges Analysis
- Customer Tenure Analysis
- Internet Service Analysis
- Payment Method Analysis

## Python Visualization



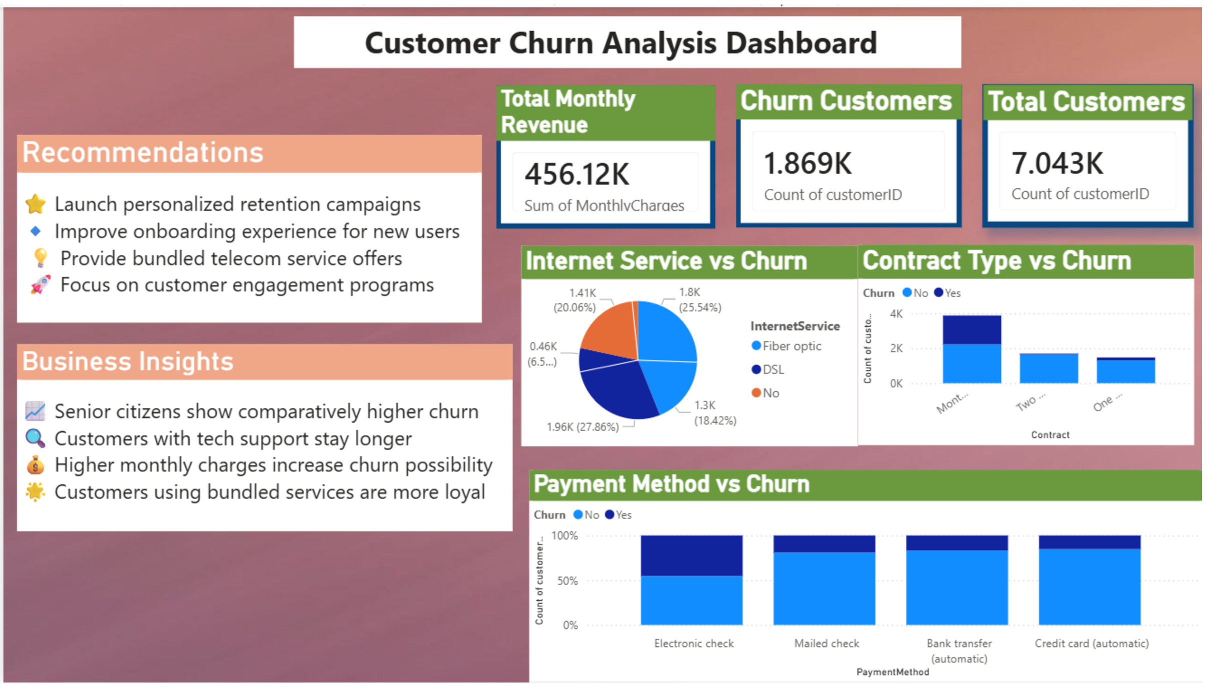

## POWER BI DASHBOARD

An interactive dashboard was developed in Power BI to visualize churn behavior and customer retention insights.

### Dashboard Components

- KPI Cards
- Revenue Overview
- Customer Churn Metrics
- Contract Analysis
- Internet Service Insights
- Payment Method Analysis
- Business Recommendations

### Dashboard Screenshot

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## MAJOR FINDINGS

- Customers with month-to-month contracts show higher churn
- Higher monthly charges increase churn probability
- Long-term customers are more likely to stay
- Customers using automatic payment methods show better retention

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## BUSINESS SUGGESTIONS

- Improve customer engagement strategies
- Encourage long-term subscription plans
- Enhance customer support services
- Introduce loyalty and retention programs

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## CONCLUSION

This project helped in understanding customer churn behavior through data analytics and visualization techniques. It also improved practical skills in Python, Power BI, and business intelligence reporting.